

LAB 3 · ARCHITECTURAL PATTERN SELECTION

Patient Health Record Consent Management System

Student: Ankita S | SRN: PES1UG24CS066

Architecture Selection: We chose Layered Architecture for the Patient Health Record Consent Management System.

Architectural style analysis

Style	Strength in this scenario	Trade-off	Fit
Layered	Clear separation of patient/clinic UI, consent rules and data access.	Minor layer-call overhead.	Best
Microservices	Independent deployment and fault isolation.	Network, deployment and data-consistency complexity can be excessive here.	Low
Client-Server	Central data control and simple clients.	Server/network dependency and availability risks.	Medium

Two scenario-specific reasons

- **1. Cohesive consent rules.** Granting or revoking time-bound consent must validate the patient, permitted record scope and expiry before access is allowed. Keeping these rules in a business layer prevents the patient and clinic interfaces from duplicating security-critical logic.
- **2. Maintainability and replaceability.** The patient portal, clinic interface and data storage can change behind stable layer boundaries. A UI or storage implementation can therefore be replaced without rewriting the core consent-validation rules.

Security advantage and performance benefit

Security advantage	Performance benefit
All health-record access passes through the business logic layer, which can enforce authentication, role-based authorization and non-expired consent before data access. The presentation components receive no direct database credentials. Parameterised repository queries and an append-only audit trail further protect integrity and accountability.	The business layer can validate consent and route an authorized record request through a predictable local workflow, avoiding unnecessary remote service calls. Indexed patient/record identifiers, prepared queries, connection pooling and appropriate caching can reduce lookup and access overhead for repeated requests.

Selected components and interfaces

Components: Patient Portal Component; Clinic Access Component; Consent Management Component; Health Record Access Component; Audit Trail Component. **Interfaces:** Consent Management Interface, Record Access Interface, Consent Verification Interface and Consent Audit Interface. The component diagram uses ball-and-socket assembly connectors and labels the interaction between components.